

Enterprise engineering studio · Intelligent software for modern commerce

Intelligent systems built to quietly run commerce.

Sanad constructs operational backbones and bespoke autonomous logic for commercial enterprises. We remove the cognitive and engineering weight of scaling — automating a structural reduction of at least 30% in daily operational load.

[Request Technical Dossier](#) →

[Read the 30% baseline](#)

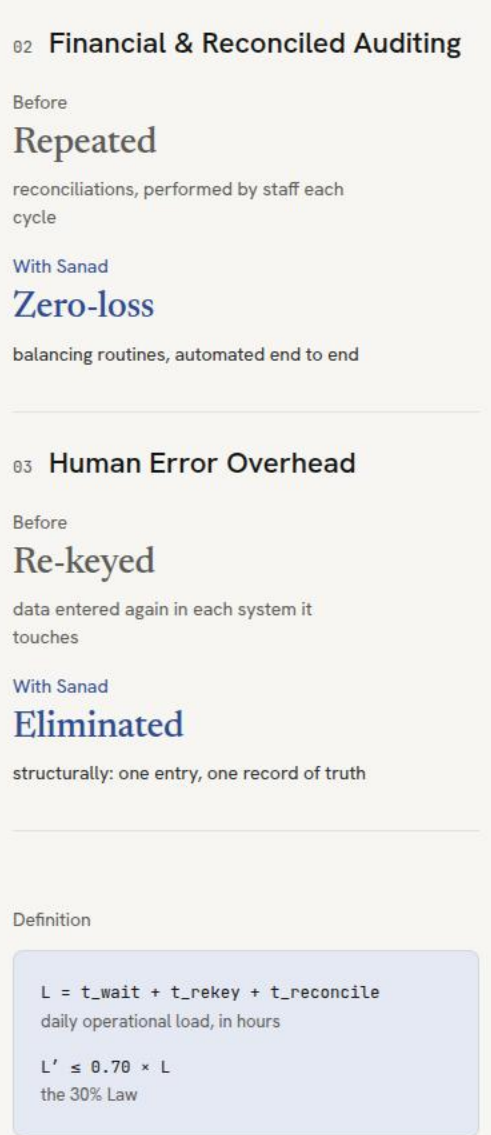
- ≥ 30%

Structural reduction in daily operational load
- 18 h → < 1 s

Inventory and supply synchronization
- < 100 ms

Internal state change across terminals
- Zero

Black-box dependencies



Enterprise data streams route into the Sanad Core modules, then synchronize out to infrastructure the client controls.

For Leadership

Predictable velocity, margin protection, and operational clarity.

For Engineering

Decoupled architecture, local-first integrity, and zero black-box dependencies.

01 Proof by metric

The 30% efficiency baseline

Operational load is not a feeling. It is waiting, re-keying and reconciling — three things that can be measured, and removed. Every system we build is specified against one baseline: a structural reduction of at least 30% in daily operational load.

01 Inventory & Supply Sync

Before

18 hours

of multi-channel lag between stock and sales

With Sanad

Sub-second

synchronization across every channel

02 Financial & Reconciled Auditing

Before

Repeated

reconciliations, performed by staff each cycle

With Sanad

Zero-loss

balancing routines, automated end to end

03 Human Error Overhead

Before

Re-keyed

data entered again in each system it touches

With Sanad

Eliminated

structurally: one entry, one record of truth

Definition

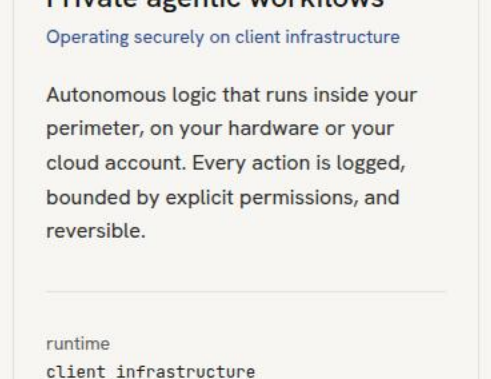
$$L = t_{\text{wait}} + t_{\text{rekey}} + t_{\text{reconcile}}$$

daily operational load, in hours

$$L' \leq 0.70 \times L$$

the 30% Law

Daily operational load, indexed baseline = 100



02 Systems & architecture

Ready-made modules. Bespoke engineering.

Two explicit offerings, never blended. Pillar A is production-tested software, configured for your business. Pillar B is engineered from scratch, for operational flows that belong to you alone.

Pre-configured SaaS · Pillar A

Turnkey Intelligent Modules

High-efficiency, production-tested systems, ready for deployment.

Select a module. Follow the data.

Choose Inventory, Units or Ledger to trace which sources feed it, which record it owns, and what it sends to your infrastructure.

- A.1

Inventory & Distributed Warehousing Logic

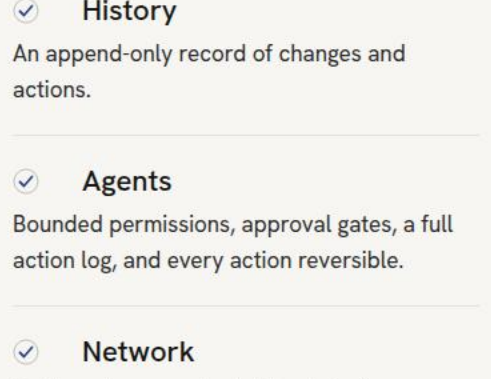
One stock position across every warehouse, store and channel.
- A.2

Hospitality & Unit Management Engines

Units, stays and turnover, held to a single source of availability.
- A.3

Unified Commerce & POS Ledger Sync

Every sale, online or in store, posts once to one ledger.



Owns

Stock position per node, reservations and transfers.

Fed by

Warehouse scanners 3PL & logistics

Schema

```
type StockLevel {
  sku      string
  node     NodeId // warehouse or store
  on_hand  int
  reserved int // held for open orders
  version  Lamport // deterministic ordering
}
```

Parameters

Topology n-node, multi-warehouse

Sync event-sourced, deterministic merge

Reservations per-channel holds, auto-expiry

Offline node-local ledger, reconcile on reconnect

Latency < 100 ms in-site, sub-second across channels

A.1 · Interface wireframe

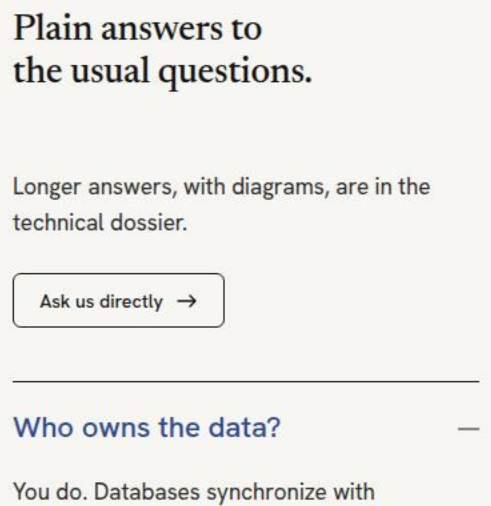
Inventory & Distributed Warehousing Logic

Stock		Transfers		Reservations		Transfer
SKU		WH-A	WH-B	WH-C		Total
SKU-1042		128	64	0		192
SKU-2210		12	40	33		85
SKU-3087		220	0	18		238
SKU-4410		7	7	7		21
SKU-5002		54	90	31		175

● Synced 0.4 s ago 5 SKUs · 3 nodes

A.2 · Interface wireframe

Hospitality & Unit Management Engines



A.3 · Interface wireframe

Unified Commerce & POS Ledger Sync

Sync	Close	Live	
POS #A-10441		Account	Dr Cr
	24.00	1100 Cash	46.75
	18.50		
		→ 4000 Sales	42.50
VAT	4.25		
Total	46.75	2200 VAT	4.25
		Σ Dr = Σ Cr	46.75

● Balanced, posted once idempotent

Bespoke systems · Pillar B

Tailored Commercial Architecture

From-scratch systems, engineered for unique, proprietary operational flows.

B.1

Custom internal operating systems

Replacing patchwork spreadsheets

One system of record in place of a folder of spreadsheets: role-based views, enforced workflows, and a history of every change.

record single source of truth

access role-based

history append-only

B.2

Private agentic workflows

Operating securely on client infrastructure

Autonomous logic that runs inside your perimeter, on your hardware or your cloud account. Every action is logged, bounded by explicit permissions, and reversible.

runtime client infrastructure

egress denied by default

oversight approval gates, full action log

B.3

High-availability distributed databases

With offline-first synchronization

Replicated stores that keep accepting writes when the network does not, then converge deterministically when it returns.

writes local-first

merge deterministic

single point of failure none, by design

03 System mechanics & technical rigor

Specified like infrastructure, stated so it can be tested.

For CTOs and technical evaluators

Data ownership

Sovereign

Complete sovereignty. Databases synchronize with client-controlled infrastructure.

Reliability

Offline-capable

Operational nodes designed to function without internet disruption.

Latency

< 100 ms

Sub-100ms internal state changes across multi-terminal setups.

Integrations

Deterministic

Clean, deterministic APIs connecting legacy ERPs, payment gateways, and third-party logistics.

Representative event trace

Illustrative sequence, not a benchmark.

```
# multi-terminal sale on the shop floor
t+000 ms sale.created tx=A-10441
t+008 ms ledger.posted ΣDr = ΣCr
t+019 ms stock.reserved sku=1042
t+037 ms state.applied v=2291
t+041 ms state.applied v=2291

# network lost: the node keeps working locally
t+912 ms sale.created tx=A-10442
t+921 ms ledger.posted queued=1

# network restored: replicate to client-db
t+4.2 s replicate converged v=2292

Target: state changes under 100 ms across terminals, with no dependency on the network.
```

04 Security & compliance

Stated plainly, so it can be checked.

Properties of the systems we build, written as specifications rather than assurances.

We do not claim certifications on this page. If you must meet a specific regulation, we design to it and show you how.

✓ **Data location**

Client-controlled infrastructure.

✓ **Access**

Role-based. Permissions are explicit and reviewable.

✓ **History**

An append-only record of changes and actions.

✓ **Agents**

Bounded permissions, approval gates, a full action log, and every action reversible.

✓ **Network**

Outbound access denied by default.

✓ **Failure**

Offline-capable nodes and deterministic recovery.

05 How we work

Four stages. One owner at each.

Every engagement follows the same order, so you always know what is being decided and what comes next.

01 Understand

We map how work actually moves through your business: systems, hand-offs, waiting and re-entry.

You receive

A written map of current operations, and the baseline for the 30% target.

02 Design

We specify the system: data model, integrations, failure behavior, and what must run offline.

You receive

An architecture document you can hand to your own engineers.

03 Build

We build and test in your environment, in stages, each one usable on its own.

You receive

Working software, tested against the baseline.

04 Operate

We run it with you, monitor it, and hand over ownership when you are ready.

You receive

Runbooks, access, and the option to run it yourselves.

06 Engagement models

Two ways to start.

Best when

Turnkey modules

Your process matches a standard shape: stock, units, a sales ledger.

Bespoke systems

Your operations are particular to you and no product fits.

What we do

Turnkey modules

Configure, connect and deploy a tested module.

Bespoke systems

Specify and engineer from scratch.

From you

Turnkey modules

Access to current systems, a named process owner, and your configuration decisions.

Bespoke systems

Time with the people who run the work, access to systems and data, and a named decision-maker.

You get

Turnkey modules

A running system, and training.

Bespoke systems

A system built to your flow, with full documentation.

Start with

Turnkey modules

The technical dossier.

Bespoke systems

An architectural walkthrough.

07 Questions from engineers

Plain answers to the usual questions.

Longer answers, with diagrams, are in the technical dossier.

[Ask us directly](#) →

Who owns the data?

You do. Databases synchronize with infrastructure you control, and we keep no copy that we depend on.

What happens without internet?

How do we migrate from our current system?

Can it connect to our ERP and payment gateway?

Do agents send our data to outside services?

What if we outgrow a turnkey module?

08 The Sanad: manifesto

Software should clarify the mind, not demand it.

Most enterprise software creates its own bureaucracy. Teams spend hours managing the tool that was supposed to save them time.

Sanad is built on the inverse thesis: intelligent software works silently in the background so leaders and builders can dedicate their focus to what actually matters.

Support for the people who lead.

If the work is heavier than it should be, we should talk.

[Request Technical Dossier](#) →

or write to inquiries@sanad.systems

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● Accepting new engagements · Remote-first studio

Documentation

Technical documentation

Whitepapers

Operational blueprints

Shared on request, with the technical dossier.

Inquiries

inquiries@sanad.systems

For an architectural walkthrough of any system on this page, write to us directly.

Support for the people who lead.